

Analytical methods for the resolution of coupled and decoupled transmon qubits

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The present work studies transmons. Here is explained the existence of two main bases for describing the eigenvalues and eigenfunctions of the system: the Number of Cooper Pair basis and the Phase basis. Firstly, is performed an analysis in the Cooper Pair basis of a single transmon, with the aid of the method of the second quantization [1] and matrix algebra, obtaining an eigenfunction that matches almost equally to the one that generates the python package Scubits [26] and eigenvalues that match almost entirely the numerical resolution of the Hamiltonian. Secondly, a study of two coupled transmons is done through a decoupling procedure that allows one to find the eigenfunctions and eigenvalues of the decoupled system. Further investigation is needed.

I. INTRODUCTION

Let us consider the wavefunction $|\hat{n}\rangle$ that describes a quantum system, we can define $|0\rangle$ as its ground state and $|1\rangle$ as its first excited state (see Ref. [18]). This defines a qubit as a mathematical object that is superposed between these two states, typically $|0\rangle$ and $|1\rangle$ in a way similar to $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$. Then, when the qubit is measured, it collapses into one of these two states. Hence, a proper definition of a qubit consists of a two-level system that can be manipulated through quantum gates represented by unitary matrices. In the present work, we will study the properties of a special kind of superconducting qubit, called the Transmon.

A. Transmon qubits

The transmon qubit has its name as an acronym of "Transmission line shunted plasma oscillation qubit" and it was first proposed by [8]. This superconducting qubit [21] allows two major advantages. Since it is a circuit, it means that we are able to bring it to desirable configurations via changing parameters, and due to their size they can be built via traditional scalable microfabrication methods. They can be considered macroscopic, implying that they will couple strongly to the surroundings, which, from one hand, allows a very good control of the system from the outside but, from the other, makes a clear entrance for noise in the system.

As mentioned in Ref. [8], the transmon qubit is based on a Cooper pair box or charge qubit, meaning that states are charge states, i.e. states which represent the presence or absence of excess Cooper pairs on an island. (For more details, see Appendix [V A]). Indeed, the Cooper pair box and the transmons [5] are based on a LC quantum circuit. Notice that a quantum LC circuit behaves similarly as a quantum oscillator. In the harmonic oscillator, energy levels are equidistant, and they are characterized by a quantum number, the integer n . Since a qubit is formed by only two states, we need to have a modified version of the harmonic oscillator in such a manner that states need to be well separated in energy. This is achieved by means of an anharmonic or non-linear terms in the Hamiltonian.

Such terms are accomplished by means of a Josephson junction.

A Josephson junction is formed [9] by separating two superconducting islands with an insulator thin enough such that Cooper Pairs are transported via quantum tunnel. Indeed, this behavior can be modeled as a pure non-linear inductance [5] in parallel with a capacitance formed by overlapping two junctions.

Performing the corresponding calculations we can derive the hamiltonian of the transmon qubit as:

$$\hat{H}_{Tr} = 4E_C (\hat{n} - n_g)^2 - E_J \cos \hat{\varphi},$$

where it is important to realize that the commutation relation (see Refs. [24], and [25]) are conjugate to each other, meaning that $[\hat{\varphi}, \hat{n}] = i\hbar$, where we see that $\hat{\varphi}$ references the phase of the qubit, and $\hat{n}$ references the number of Cooper Pairs. From here we will take $\hbar = 1$.

II. SINGLE TRANSMON

As we have previously discussed, the transmon qubit is an evolution of the Cooper-pair box qubit, both of which are described by similar Hamiltonians. The primary distinction between the two lies in the ratio of the Josephson energy (E_J) to the charging energy (E_C).

In the transmon design, the Hamiltonian remains largely the same as that of the CPB qubit, but with a significant difference in the $\frac{E_J}{E_C}$ ratio. Here, E_J is the Josephson energy of the junction, representing the energy scale associated with the tunneling of Cooper pairs through the Josephson junction, and $E_C = \frac{e^2}{2C}$ is the charging energy, which is inversely proportional to the total capacitance (C) of the qubit circuit.

For transmons, the $\frac{E_J}{E_C}$ ratio is typically much greater than one ($\frac{E_J}{E_C} \gg 1$), contrasting with the CPB qubit where this ratio is usually less than or around one ($\frac{E_J}{E_C} \leq 1$). This significant difference in ratio is achieved by shunting the Josephson junction with an additional large capacitor, thereby increasing the total capacitance and reducing the charging energy.

The consequence of this high $\frac{E_J}{E_C}$ ratio in transmons is a substantial reduction in the sensitivity of the qubit's

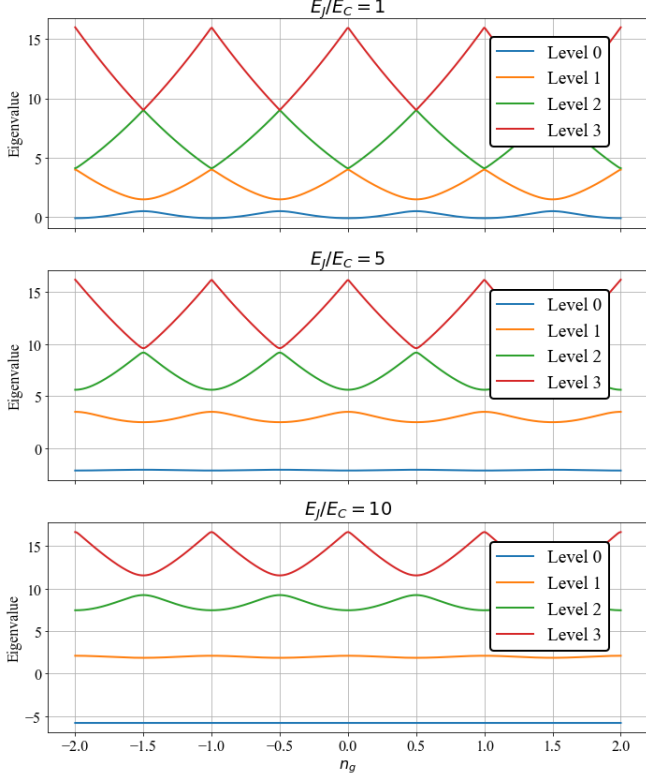


Рис. 1: Transmon energy spectrum for small values of the ratio $\frac{E_J}{E_C}$ as indicated.

energy levels to charge noise, making the transmon qubit more robust against decoherence mechanisms related to charge fluctuations. This enhancement in stability is a key factor that has made transmons a popular choice in the field of superconducting qubits.

In the following, we are going to solve the Hamiltonian of the transmon in order to obtain the eigenvectors and eigenvalues of such system. Depending on the ratio $\frac{E_J}{E_C}$ the system spectrum behaves as a Cooper pair box ($\frac{E_J}{E_C} = 1$) or as a transmon ($\frac{E_J}{E_C} = 10$). Looking to more dramatic values for this ratio (from $\frac{E_J}{E_C} = 50$ to $\frac{E_J}{E_C} = 100$) we can see how the spectrum resembles the harmonic oscillator spectrum with some anharmonicity component.

In the following we show how we have computed the eigenvalues and the corresponding eigenstates by using the Cooper pair representation.

A. Canonical transformation for single transmon

The Hamiltonian reads

$$\hat{H}_{Tr} = 4E_C (\hat{n} - n_g)^2 - E_J \cos(\hat{\varphi}).$$

By defining the variables η and w as $\eta = \sqrt{\frac{E_C}{8E_J}}$ and $w = \sqrt{8E_J E_C}$ respectively, as is shown in the Appendix

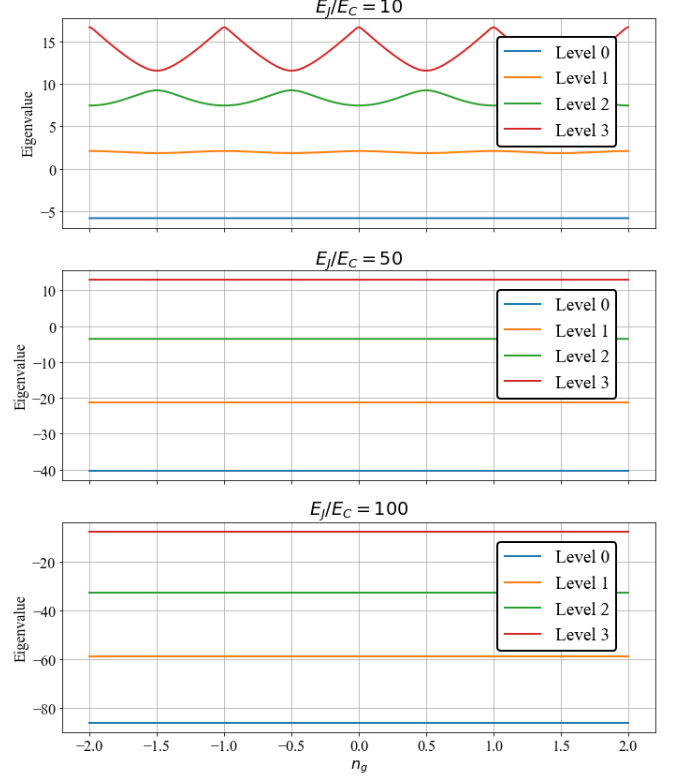


Рис. 2: Transmon energy spectrum for large values of the ratio $\frac{E_J}{E_C}$ as indicated.

$[VD]$, we can simplify $\hat{H}_{Tr}$ as

$$\frac{\hat{H}_{Tr}}{w} = 4\eta (\hat{n} - n_g)^2 - \frac{1}{8\eta} \cos(\hat{\varphi}), \quad (1)$$

and together with the canonical transformation between the creation and annihilation operators listed as

$$\hat{\varphi} = 2\sqrt{\eta} (\hat{a} + \hat{a}^\dagger) \quad \hat{n} - n_g = \frac{(\hat{a} - \hat{a}^\dagger)}{4i\sqrt{\eta}}. \quad (2)$$

If we insert these relations into the Hamiltonian it can be written as

$$H = \frac{1}{4} (a - a^\dagger)^2 - \frac{1}{8\eta} \cos(2\sqrt{\eta}(a + a^\dagger)).$$

Notice that for $\eta \ll 1$ one can make a Taylor expansion for the cosine function considering only the first terms to describe the transmon limit $E_C \gg E_J$. However, in the next section we obtain the eigenfunctions and eigenvalues for arbitrary values of this ratio.

B. Eigenfunctions and vacuum vector

We have an automorphism that produces a canonical transformation that will be a proper one because we have a finite number of degrees of freedom. We know from Ref. [1] that there is a vacuum vector, F_0 , in the state space (see Appendix [V B]), such that $\hat{a}F_0 = 0$. This is precisely our method. We are going to set $\hat{a}F_0 = 0$ to obtain the wavefunction for the ground state. Excited states will be achieved by the application of the ladder operators $a^\dagger$ as many times as needed it. From Eq. (26) we can isolate a and $a^\dagger$ as a function of $\hat{n}$ and $\hat{\varphi}$:

$$\begin{aligned}\hat{a} &= \left(i2\sqrt{\eta}(\hat{n} - n_g) + \frac{\hat{\varphi}}{4\sqrt{\eta}} \right) \\ \hat{a}^\dagger &= \left(-i2\sqrt{\eta}(\hat{n} - n_g) + \frac{\hat{\varphi}}{4\sqrt{\eta}} \right).\end{aligned}\quad (3)$$

Identifying the vacuum vector of our system as $F_0 = |0\rangle$ we have $\hat{a}|0\rangle = 0$. Which is translated into

$$\hat{a}|0\rangle = \left(i2\sqrt{\eta}(\hat{n} - n_g) + \frac{\hat{\varphi}}{4\sqrt{\eta}} \right) |0\rangle = 0. \quad (4)$$

With the aid of Appendix [V E], and the commutation relation $[\hat{\varphi}, \hat{n}] = i$ that implies $[\hat{a}, \hat{a}^\dagger] = 1$, we arrive at

$$\hat{\varphi}\hat{F} = i\frac{\delta}{\delta n}\hat{F} \quad \hat{n}\hat{F} = n\hat{F}. \quad (5)$$

Reorganizing the previous expression, the ground state wavefunction in terms of the Cooper pair n variable fulfills the following differential equation

$$(8\eta(n - n_g) + \partial_n)|0\rangle = 0. \quad (6)$$

Solving the equation by standard procedures the solution is

$$\langle n|0\rangle = \left(\frac{8\eta}{\pi} \right)^{\frac{1}{4}} e^{-4\eta(n - n_g)^2} \quad (7)$$

Now we present our results. We plot $|\langle n|0\rangle|^2$ in Fig. (3) to see the decomposition of the ground state. What is really interesting is that we can work with a Python package called ScQubits [26] and make a plot with the same parameters as in Fig. (4).

Comparing both Fig. (26) and Fig. (26), we see a good agreement between them. In our graphics, we have a state decomposition in the quantum numbers $n = \{-3, -2, -1, 0, 1, 2, 3\}$, where the only nonzero values of the wavefunction are based on $n = \{-1, 0, 1, 2\}$ for both. As we can also observe, the amplitude is in quite good agreement. However, there is a difference between the

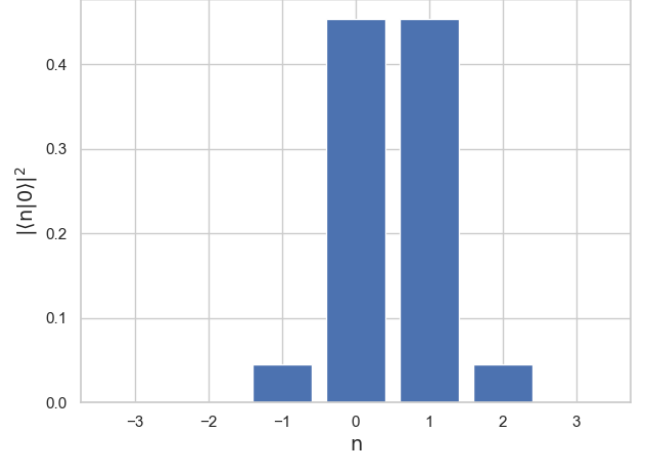


Рис. 3: Analytical wavefunction in the $\hat{n}$ representation, see Eq. (7). The used parameters are: $E_C = 6$, $E_J = 36$, $n_g = \frac{1}{2}$.

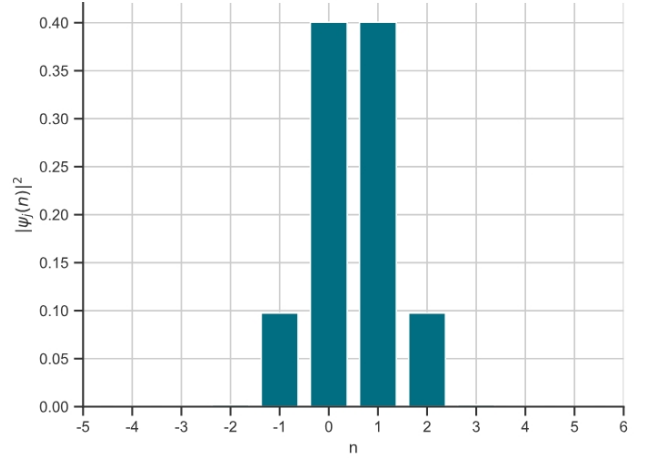


Рис. 4: Numerical Fourier transform of the wavefunction resolved in the φ basis by using Scqubit code. The used parameters are: $E_C = 6$, $E_J = 36$, $n_g = \frac{1}{2}$.

graphics. As we may see, the amplitude of the states with $n = \{-1, 2\}$ is lower in our function and the states with $n = \{0, 1\}$ is higher. One may wonder why this happens. At first sight, we see that the state decomposition of the Scqubits package comes from a Fourier transform of the adapted corresponding Mathieu functions, but what really lies beyond this is the fact that our function is narrower than the one that produces Scqubits. This fact only implies that the probability is a bit sifted towards either having 0 Cooper Pairs that crossed or one.

C. Eigenvalues

In this section, we apply a method to obtain the transmon spectrum in the Cooper pair representation basis. We refer to Appendices [V F] and [10] for details.

The transmon Hamiltonian elements $H_{Tr;l,j} = \langle l | H_{Tr} | j \rangle$ in the charge representation read as

$$H_{Tr;l,j} = 4E_c(l - n_g)^2 - \frac{E_J}{2}(\delta_{l+1,j} + \delta_{l,j+1}). \quad (8)$$

In order to make progress analytically, we assume a truncation scheme and consider only a 4×4 matrix, that is, $\dim(H_{Tr,l,j}) = 4$.

We use here a method consisting in the construction of a special kind of continued fraction by a successive dianogalization. We refer the reader to the Appendix [V F] for details. In brief, the method allows us to iteratively construct a series of matrices named L_j and R_j that allow, via a product with the Hamiltonian Matrix, to isolate the diagonal element by making zero all the components of the corresponding row and column except the one in the diagonal. The basic procedure of this method consists of making similarity transformations in order to arrive at a triangular matrix, that, when done in the upper and lower region, a diagonal form is obtained. This method begins with $H_{Tr,11}$ and ends performing a transformation in the $H_{Tr,l,j}$ element, being this the one that accumulated all the previous information and all the fractions needed for the lower domain. By construction of our matrix, we are interested in the element $H_{Tr,11}$. For having all the information accumulated, there is needed to be the last transformed element. For performing this, a modification of the method is needed. The modifications are as follows:

1. Start the diagonalization from the lowest row and column, isolating the element $H_{Tr,l,j}$, in our case $H_{Tr,44}$ and end in the element $H_{Tr,11}$.
2. The order of application of the matrices has to be changed in order to get the desired isolation of each element, for this, a symmetry of application is seen, meaning that a transformation $L_i \rightarrow R_i$ and $R_i \rightarrow L_i$ has to be performed. This is translated to simply say that what was the L matrix is now the R matrix and vice versa.

Following the procedure, we can define the following variables:

$$\begin{aligned} r_1 &= -\frac{-\frac{E_J}{2}}{4E_c(n+3-n_g)^2} \\ r_2 &= -\frac{-\frac{E_J}{2}}{4E_c(n+2-n_g)^2 - r_1} \\ r_3 &= -\frac{-\frac{E_J}{2}}{4E_c(n+1-n_g)^2 - r_2} \end{aligned}$$

that will allow us to obtain the following R and L associated matrices

$$R_1 = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & r_1 & 1 \end{pmatrix} \quad L_1 = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & r_1 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

$$R_2 = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & r_2 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \quad L_2 = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & r_2 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

$$R_3 = \begin{pmatrix} 1 & 0 & 0 & 0 \\ r_3 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \quad L_3 = \begin{pmatrix} 1 & r_3 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}.$$

Using the previous results we can directly write down

$$D_{H_{Tr}} = (L_3 (L_2 (L_1 H_{Tr} R_1) R_2) R_3) = \text{diag}\{\lambda_n, \lambda_{n+1}, \lambda_{n+2}, \lambda_{n+3}\} \quad (9)$$

and hence gets the definition for any given state $\{\hat{n}\}$ as

$$\lambda(n \geq 1) = 4E_c(n - n_g)^2 - \frac{E_J^2}{4 \left(4E_c(n - n_g + 1)^2 - \frac{E_J^2}{4 \left(4E_c(n - n_g + 2)^2 - \frac{E_J^2}{16E_c(n - n_g + 3)^2} \right)} \right)}. \quad (10)$$

For a reason that needs further investigation when $n = 0$ we do not get the right behavior. We are able to obtain a better approximation for the fundamental eigenvalue if we do as follows.

If we approximate by Taylor expansion the term $\cos \phi$, we get $\cos \phi \approx 1 + \frac{1}{2} \frac{\partial^2}{\partial n^2}$. Calculating $\langle 0 | \hat{H}_{Tr} | 0 \rangle$, keeping in mind that $\hat{n}|n\rangle = n|n\rangle$ we can write

$$\lambda(n = 0) = \left(\frac{1}{2} - \frac{1}{8\eta} \right) w, \quad (11)$$

where we apply the definitions of η and w given above.

We now check the validity of this expressions. We construct a matrix representation for the Hamiltonian and then we obtain a numerical solution to compare with. Both results are plotted in Fig. (5).

As Fig. (5) shows, we have a good agreement between the analytical functions and the numerical solutions. For the $n = 1$ states, we observe, at the very beginning of the domain, exactly 4 infinities, that are cut for aesthetic purposes, which are due to the 3 discontinuities of the analytical function, but once these are surpassed, we can see a perfect alignment between the analytical

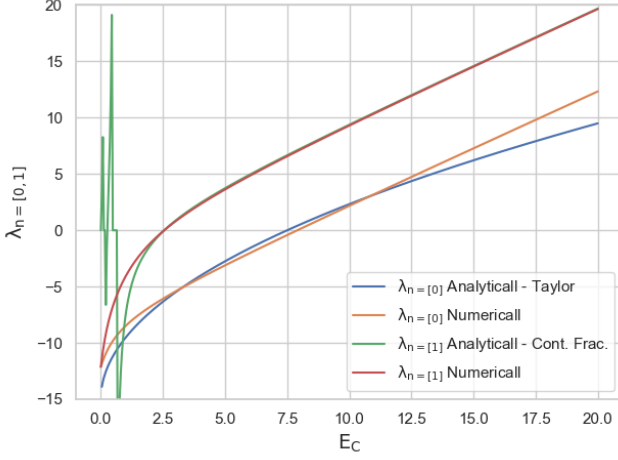


FIG. 5: Comparison of the analytically found general eigenvalue [10] for $\hat{n} = [0, 1]$ with the proper numerical resolution of the transmon Hamiltonian. The parameters are: $E_C = [0, 20]$, $E_J = 15$, $\hat{n} = [0, 1]$, $n_g = \frac{1}{2}$.

and numerical eigenvalues. This might be caused by the method used to diagonalize the matrix, since it gives a continued fraction. For the plot of the eigenvalues with $n = 0$ things start to get a bit strange. First of all, when we use the, what appears to be, general formula 10, we get a strange behavior, since neither the origin, the slope, nor the values resemble the numerical solution. This needs further investigation. To solve this problem, we should attack directly the hamiltonian and perform the expected value with the states $|0\rangle$ and apply a Taylor decomposition for the operator $\cos(\varphi)$ until $O(3)$, then we obtain 11 and plot it with the numerical solution. This, as we can see, is not as accurate as the $n = 1$ plot, but follows the pattern for a given domain. One possible solution for this domain is to change the regime in which we perform the Taylor expansion. Here, further investigation is also needed.

If the reader is interested in a perturbative analysis applied to the eigenvalues, we recommend Refs. [9] and [21].

III. TWO COUPLED TRANSMONS

Here we extend our study to the two-transmon system. The problem is interesting by itself as long as two-qubit operations are needed for a complete scheme for quantum computation and quantum algorithms, needing two qubit gates, which can be done in several ways. One of these is through tunable coupling, achieved by two capacitatively coupled transmons, presenting the advantage of having more coherence [32].

Considering a circuit of two capacitatively coupled transmons, we can write the corresponding Hamiltonian consisting of three main terms, two of them corresponding to the single transmons and the last

one representing the coupling between them. Following Ref. [11] we write

$$H_{Tr;c} = \sum_{i=1,2} \left[4E_{C_i} (n_i - n_{g_i})^2 - E_{J_i} \cos \varphi_i \right] + 4E_I n_1 n_2,$$

that translates into

$$H_{Tr;c} = 4E_{C_1} (n_1 - n_{g_1})^2 + 4E_{C_2} (n_2 - n_{g_2})^2 + 4E_I n_1 n_2 - E_{J_1} \cos \varphi_1 + E_{J_2} \cos \varphi_2. \quad (12)$$

As in Ref. [13], let us suppose that $\exists \pi : (E \times E, F \times F) \rightarrow \text{Im}(\pi)$, then π is the coupling of μ, ν if it verifies that

$$\pi(A \times E) = \mu(A) \quad \pi(E \times A) = \nu(A).$$

This means that $\pi : E \times E \rightarrow E$ is a bilinear automorphism, meaning that π takes two arguments from E and gives one from E . This, together with the conditions that must verify, can be achieved by the product measure, leaving us with $\pi = \mu \times \nu$, but given that we are in a topological space, we can define the coupling as

$$\pi = \nu\mu = \text{coupling}.$$

Specifically, for our problem, we will have $\pi = \hat{n}_1 \hat{n}_2$ and, renormalizing, we will end up with

$$\pi = 4E_I \hat{n}_1 \hat{n}_2.$$

As seen, we can define the coupling as the product of spaces, meaning the product of variables. This leads to the natural definition of uncoupling as separating the product. We know that $\forall \hat{B} : D \subset \mathbb{L} \rightarrow \mathbb{L} \exists \hat{U} : \hat{B} = \hat{U} \hat{B} \hat{U}^{-1}$, meaning that it will exist an automorphism that we can use to uncouple the variables. Let us put in correspondence the variables $\{\hat{n}_1, \hat{n}_2\} \rightarrow \{\hat{s}_1, \hat{s}_2\}$. Then we perform a transformation of the form

$$\hat{n}_1 = \alpha \hat{s}_1 + \beta \hat{s}_2$$

$$\hat{n}_2 = \gamma \hat{s}_1 + \epsilon \hat{s}_2,$$

with the additional requirement that their commutators vanish, i.e., $[\hat{n}_{1,2}, \hat{n}_{1,2}] = 0 = [\hat{n}_{2,1}, \hat{n}_{1,2}]$ because the variables $\hat{n}_1$ and $\hat{n}_2$ belong to different subspaces. This requirement is translated in a determined choice of the coefficients together with the request of having a transformation to be unitary. This will lead us to the existence of a transformation group (see Appendix [VG]). The algebra of the so called $O^+(2)$ group, that

allows a representation of the form of Eq. (32), which is indeed a rotation, meaning that

$$\begin{pmatrix} \hat{s}_1 \\ \hat{s}_2 \end{pmatrix} = \begin{pmatrix} \cos(\alpha) & -\sin(\alpha) \\ \sin(\alpha) & \cos(\alpha) \end{pmatrix} \begin{pmatrix} \hat{n}_1 \\ \hat{n}_2 \end{pmatrix} \equiv \begin{pmatrix} \hat{n}_1 \\ \hat{n}_2 \end{pmatrix} = \begin{pmatrix} \cos(\alpha) & \sin(\alpha) \\ -\sin(\alpha) & \cos(\alpha) \end{pmatrix} \begin{pmatrix} \hat{s}_1 \\ \hat{s}_2 \end{pmatrix}. \quad (13)$$

Applying this transformation, and defining the following new variables as

$$\tilde{s}_k = s_k \sqrt{\frac{A_k}{B_k}} \quad \tilde{n}_{g_k} = n_{g_k} \sqrt{\frac{\epsilon_k}{B_k}},$$

being $\epsilon_k = (E_{C_1}, E_{C_2})$ and where

$$\begin{aligned} A_1 &= E_{C_1} \cos^2 \alpha + E_{C_2} \sin^2 \alpha - E_I \cos \alpha \sin \alpha \\ B_1 &= E_{C_2} \sin \alpha \frac{n_{g_2}}{n_{g_1}} - E_{C_1} \cos \alpha \end{aligned} \quad (14)$$

$$\begin{aligned} A_2 &= E_{C_1} \sin^2 \alpha + E_{C_2} \cos^2 \alpha + E_I \sin \alpha \cos \alpha \\ B_2 &= -E_{C_1} \sin \alpha \frac{n_{g_1}}{n_{g_2}} - E_{C_2} \cos \alpha, \end{aligned} \quad (15)$$

the following restrictions must be fulfilled

$$\begin{aligned} E_{C_2}^3 \left(\frac{n_{g_2}}{n_{g_1}} \right)^2 + E_{C_2}^2 E_{C_1} \left(1 + 2 \frac{n_{g_2}}{n_{g_1}} \right) + \\ E_{C_2} E_{C_1}^2 \left(1 - 2 \frac{n_{g_1}}{n_{g_2}} \right) - E_{C_1}^3 \left(\frac{n_{g_1}}{n_{g_2}} \right)^2 &= 0; \\ E_I &= \left| \frac{E_{C_1} - E_{C_2}}{3} \right|. \end{aligned} \quad (16)$$

Together with the requirement that α is

$$\alpha = \arctan \left[\left| \frac{E_{C_1} - E_{C_2}}{E_I} \right| + \sqrt{\left| \frac{E_{C_1} - E_{C_2}}{E_I} + 1 \right|} \right].$$

This transformation for the special choice of parameters that fulfills Eq. (16) leads to a system of two uncouple transmons. The transformed Hamiltonian read as

$$\begin{aligned} \tilde{H}_{Tr} &= \left[B_1 (\tilde{s}_1 - \tilde{n}_{g_1})^2 - E_{J_1} \cos \varphi_1 \right] \\ &+ \left[B_2 (\tilde{s}_2 - \tilde{n}_{g_2})^2 - E_{J_2} \cos \varphi_2 \right]. \end{aligned} \quad (17)$$

A. Eigenvectors and vacuum state

Given the form of the transformed Hamiltonian [17] we can express now its eigenvectors as product of

single transmon wavefunctions (see Appendix [V H] for more details). Which will imply that $\hat{a}_1|0_1\rangle = 0$ and $\hat{a}_2|0_2\rangle = 0$, and hence we will have only to care about solving a general equation $\hat{a}_k|0_k\rangle = 0$ and construct the wavefunction of the coupled system as shown.

Once again, we are dealing with an automorphism, where having in mind the Appendix [V D], we know that our variable set $\{\hat{s}_1, \hat{s}_2, \hat{\varphi}_1, \hat{\varphi}_2\}$, will take the form

$$(\tilde{s}_{1,2} - \tilde{n}_{g_{1,2}}) = \alpha_{1,2} \hat{a}_{1,2} + \beta_{1,2} \hat{a}_{1,2}^\dagger$$

$$\hat{\varphi}_{1,2} = \gamma_{1,2} \hat{a}_{1,2} + \epsilon_{1,2} \hat{a}_{1,2}^\dagger$$

and must verify, the commutation relations that follow

$$[\hat{\varphi}_{1,2}, \hat{\varphi}_{1,2}] = 0 = [\tilde{s}_{1,2}, \tilde{s}_{1,2}] \quad [\hat{\varphi}_{1,2}, \tilde{s}_{1,2}] = i\delta_{1,2}^{1,2}.$$

If we can define the variables

$$\eta_k = \sqrt{\frac{B_k}{8E_{J_k}}} \quad w_k = \sqrt{8B_k E_{J_k}},$$

operating, we find the same canonical transformation. Defining

$$\tilde{B}_{1,2} = 2\sqrt{\frac{\eta_{1,2}}{w_{1,2}}} \quad \tilde{A}_{1,2} = \frac{1}{4i\sqrt{\eta_{1,2}w_{1,2}}}$$

we are able to write that the canonical transformation takes the form

$$\begin{pmatrix} \hat{\varphi}_{1,2} \\ (\tilde{s}_{1,2} - \tilde{n}_{g_{1,2}}) \end{pmatrix} = \begin{pmatrix} \tilde{B}_{1,2} & \tilde{B}_{1,2} \\ \tilde{A}_{1,2} & -\tilde{A}_{1,2} \end{pmatrix} \begin{pmatrix} \hat{a}_{1,2} \\ \hat{a}_{1,2}^\dagger \end{pmatrix},$$

letting us write that

$$\hat{a}_{1,2} = \left(2i\sqrt{\eta_{1,2}w_{1,2}} (\tilde{s}_{1,2} - \tilde{n}_{g_{1,2}}) + \frac{\varphi_{1,2}}{4i} \sqrt{\frac{w_{1,2}}{\eta_{1,2}}} \right). \quad (18)$$

This will imply that $\hat{a}_1|0_1\rangle = 0$ and $\hat{a}_2|0_2\rangle = 0$, and hence we will have

$$\hat{a}_{1,2}|0_{1,2}\rangle = [8\eta_{1,2} (\tilde{s}_{1,2} - \tilde{n}_{g_{1,2}}) - \partial_{n_{1,2}}] |0_{1,2}\rangle = 0.$$

Now, let us redefine, in order to get back the variable $\hat{s}_k$

$$\tilde{\eta}_k = \sqrt{\frac{B_k}{8E_{J_k}}} \rightarrow \tilde{\nu}_k = \tilde{\eta}_k \sqrt{\frac{A_k}{B_k}},$$

which will imply a redefinition of the variable $\tilde{n}_{g_k}$ as

$$N_{g_k} = \tilde{n}_{g_k} \sqrt{\frac{B_k}{A_k}}$$

enabling us to write

$$[8\tilde{\nu}_{1,2}(s_{1,2} - N_{g_{1,2}}) - \partial_{n_{1,2}}] |0_{1,2}\rangle = 0,$$

which, as we know from the previous calculation, has a solution of the form

$$\langle s_k | 0_k \rangle = \left(\frac{8\tilde{\nu}_k}{\pi} \right)^{\frac{1}{4}} e^{-4\tilde{\nu}_k(s_k - N_{g_k})^2}$$

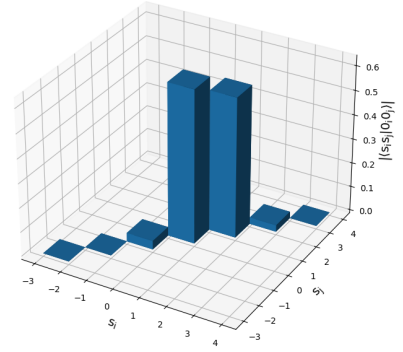
that enables us to write the general solution as

$$\langle s_1 s_2 | 0_1 0_2 \rangle = \sqrt{\frac{16}{\pi}} (\tilde{\nu}_1 \tilde{\nu}_2)^{\frac{1}{4}} e^{-4(\tilde{\nu}_1(s_1 - N_{g_1})^2 + \tilde{\nu}_2(s_2 - N_{g_2})^2)}, \quad (19)$$

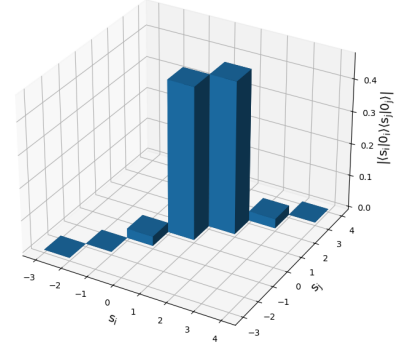
Since we have obtained the fundamental wavefunction for the coupled transmon system, it is interesting to see its behavior to compare with two independent transmon wavefunctions. In order for doing this, as we have demonstrated earlier, we compared the exact wavefunction obtained with a naive wavefunction built from the product of two transmons wavefunction in terms of the bare parameters of the two transmon Hamiltonian.

In Fig. (6) and in Fig. (7) we, as mentioned, plot in the upper part the coupled-decoupled wavefunction and in the lower part the naively constructed wavefunction. In Fig. (6), there are the wavefunctions under the condition $E_{C_1} \leq \{E_{J_1}, E_{J_2}\}$, and Fig. (7), there are the wavefunctions under the condition of $E_{C_1} \geq \{E_{J_1}, E_{J_2}\}$. Both representations verify conditions $E_i \leq \{E_{C_1}, E_{C_2}, E_{J_1}, E_{J_2}\}$ and $E_{C_2} \leq E_{C_1}$.

Under these circumstances, we see, at first glance, that the effect of the coupling, as seen in Fig. (6a) and Fig. (7a) compared to Fig. (6b) and Fig. (7b), or even Fig. (26), is to apply a rupture of the symmetry to the probability of state decomposition. As in both Fig. (6a) and Fig. (7a) we see a higher probability for the $\{s_1 = 0, s_2 = 0\}$ while the one corresponding to $\{s_1 = 1, s_2 = 1\}$ is slightly lower. This implies that the coupling prioritizes, in a higher or lower measure, one configuration. The question that arises here is "Why?" If we take a look at the expression we are plotting, Eq. 19, we see that we have a normalized negative exponential, but a closer look might tell us that the expression must be symmetrical, at least for $\{s_1 = 0, s_2 = 0\}$ and $\{s_1 = 1, s_2 = 1\}$, but if we look more carefully we see that we have N_{g_1} and N_{g_2} which have a dependence on B_k and A_k , and if we look at the expressions Eqs. 14 and 15 we see the action of the coupling energy E_i , which ultimately affects the expression of N_{g_k} . This directly translates into the asymmetry that we observe.



(a) Found analytical wavefunction (19) of the coupled transmons. Here $E_{C_1} = 10$, $E_{J_1} = 36$, $E_{J_2} = 20$, $E_{C_2} = 4.14$ and $E_I = 1.95$.



(b) Direct product of the wavefunctions (7) for comparing with the coupled case. Here $E_{C_1} = 10$, $E_{J_1} = 36$, $E_{J_2} = 20$ and $E_{C_2} = 4.14$

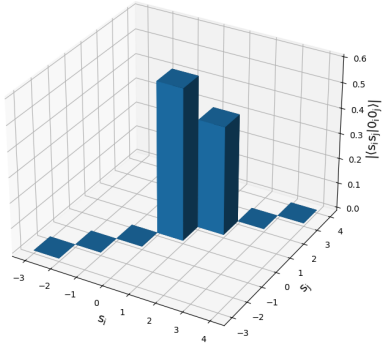
FIG. 6: 3D bar plot of the coupled transmon eigenfunction (19) and the direct product of the single transmon eigenfunction (7) in order to get a comparison of the behavior of both of them. The parameters are: $s_{1,min} = -3$, $s_{2,min} = -3$, $n_{g_1} = \frac{1}{2}$, $n_{g_2} = \frac{1}{2}$.

We are clear to think that, precisely, the paper of the coupling is to provoke a rupture of the symmetry due to the interaction between the transmons.

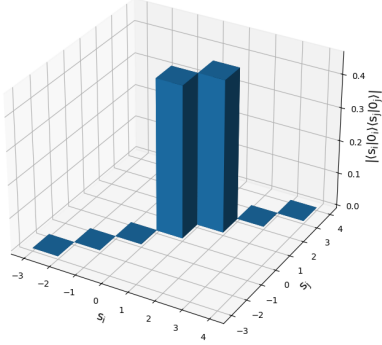
Here we can visualize two more things. The first is that the higher E_{C_1} , the higher the asymmetry. The second and most important point is that the naively constructed wavefunction to compare cannot reproduce this asymmetry. Showing us why is important the wavefunction that we have gotten.

B. Eigenvalues

Given the form of the transformed Hamiltonian, $\hat{H}_{Tr}$ (17), we can translate it to its matrix representation:



(a) Found analytical wavefunction (19) of the coupled transmons. Here $E_{C1} = 30$, $E_{J1} = 26$, $E_{J2} = 10$, $E_{C2} = 12.43$ and $E_I = 5, 86$.



(b) Direct product of the wavefunctions (7) for comparing with the coupled case. Here $E_{C1} = 30$, $E_{J1} = 26$, $E_{J2} = 10$ and $E_{C2} = 12.43$.

Рис. 7: 3D bar plot of the coupled transmon eigenfunction (19) and the direct product of the single transmon eigenfunction (7) in order to get a comparison of the behavior of both of them. The parameters are: $s_{1,min} = -3$, $s_{2,min} = -3$, $n_{g1} = \frac{1}{2}$, $n_{g2} = \frac{1}{2}$.

$$\tilde{H}_{Tr_{ij;pq}} = \left[A_1 (l - N_{g1})^2 - \frac{E_{J1}}{2} (\delta_{l+1,j} + \delta_{l,j+1}) \right] + \left[A_2 (p - N_{g2})^2 - \frac{E_{J2}}{2} (\delta_{p+1,q} + \delta_{p,q+1}) \right].$$

Now, we have defined $\hat{H}_{Tr} : \mathbb{L} \rightarrow \mathbb{L}$, but, as we see, we are dealing with an uncoupled system, meaning that we can define $H_1 \subset \mathbb{L}$ and $H_2 \subset \mathbb{L}$ such that $H_1 \cap H_2 = \emptyset$ where we have that $\hat{H}_{Tr;1} \in H_1$ and $\hat{H}_{Tr;2} \in H_2$. This allows us to write $\hat{H}_{Tr} = \hat{H}_{Tr;1} + \hat{H}_{Tr;2}$.

Given this, if we define $sp(\hat{H}_{Tr})$ as the spectrum of our operator, given the above argument, we will have that $sp(\hat{H}_{Tr}) = sp(\hat{H}_{Tr;1}) + sp(\hat{H}_{Tr;2})$. And hence, we will have the following functions

$$\lambda_{dc}(\tilde{s}_1 = 0, \tilde{s}_2 = 0) = \left(\frac{1}{2} - \frac{1}{8\tilde{\nu}_1} \right) \omega_1 + \left(\frac{1}{2} - \frac{1}{8\tilde{\nu}_2} \right) \omega_2 \quad (20)$$

$$\lambda_{dc}(\tilde{s}_1 \geq 1, \tilde{s}_2 = 0) = A_1 (s_1 - N_{g1})^2 - \frac{E_{J1}^2}{4 \left(A_1 (s_1 - N_{g1} + 1)^2 - \frac{E_{J1}^2}{4 \left(A_1 (s_1 - N_{g1} + 2)^2 - \frac{E_{J1}^2}{4 A_1 (s_1 - N_{g1} + 3)^2} \right)} \right)} + \left(\frac{1}{2} - \frac{1}{8\tilde{\nu}_2} \right) \omega_2 \quad (21)$$

$$\lambda_{dc}(\tilde{s}_1 = 0, \tilde{s}_2 \geq 1) = \left(\frac{1}{2} - \frac{1}{8\tilde{\nu}_1} \right) \omega_1 + A_2 (s_2 - N_{g2})^2 - \frac{E_{J2}^2}{4 \left(A_2 (s_2 - N_{g2} + 1)^2 - \frac{E_{J2}^2}{4 \left(A_2 (s_2 - N_{g2} + 2)^2 - \frac{E_{J2}^2}{4 A_2 (s_2 - N_{g2} + 3)^2} \right)} \right)} \quad (22)$$

$$\lambda_{dc}(\tilde{s}_1 \geq 1, \tilde{s}_2 \geq 1) = A_1 (s_1 - N_{g1})^2 - \frac{E_{J1}^2}{4 \left(A_1 (s_1 - N_{g1} + 1)^2 - \frac{E_{J1}^2}{4 \left(A_1 (s_1 - N_{g1} + 2)^2 - \frac{E_{J1}^2}{4 A_1 (s_1 - N_{g1} + 3)^2} \right)} \right)} + A_2 (s_2 - N_{g2})^2 - \frac{E_{J2}^2}{4 \left(A_2 (s_2 - N_{g2} + 1)^2 - \frac{E_{J2}^2}{4 \left(A_2 (s_2 - N_{g2} + 2)^2 - \frac{E_{J2}^2}{4 A_2 (s_2 - N_{g2} + 3)^2} \right)} \right)} \quad (23)$$

for any given state of $\{\tilde{s}_1, \tilde{s}_2\}$.

Once we have the corresponding functions that represent the eigenvalues for the decoupled transmon, as seen in Eqs. (20), (21), (22) and (23), it is interesting to plot and analyze the behavior of these eigenvalues. Here we cannot compare them to a numerical solution because it would require tensors, and it is beyond our current analysis. There are some interesting features that must be acknowledged.

If we take a closer look at our plot, Fig. (??), we can see that the behavior, meaning the form of the curve, resembles the single transmon, as seen in Fig. (5), which might lead us to think that our calculations are in good agreement with reality. Once again, as before, there are three discontinuities due to the construction of the eigenvalues, which have been cut for aesthetic purposes.

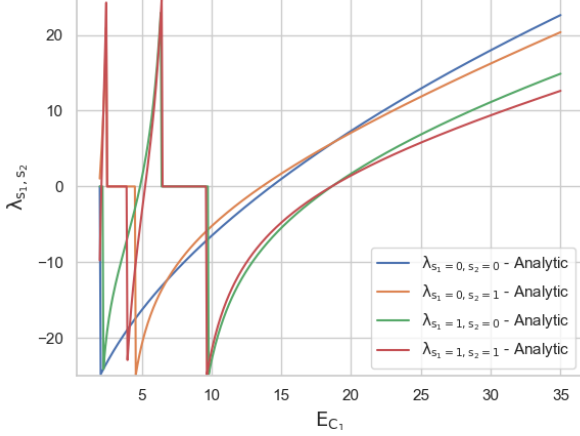


Рис. 8: Plot of the analytically found eigenvalues (??) for the coupled system for the four possible combinations of the lowest values of $\hat{s}_1$ and $\hat{s}_2$. The parameters are:
 $\{\hat{s}_1, \hat{s}_2\} = \{[0, 1], [0, 1]\}$, $E_{J_1} = 15$, $E_{J_2} = 25$, $n_{g_1} = \frac{1}{2}$,
 $n_{g_2} = \frac{1}{2}$.

We see that, in comparison to the single transmon, the umbral where the eigenvalues begin to behave normally is higher, which might be caused due to fluctuations in the fractions.

In addition to these facts, here arises a phenomenon that was not expected. We can clearly see the superposition or overlap of two and two functions. In particular, we see that the eigenvalues for $\{s_1 = 0, s_2 = 0\}$ and $\{s_1 = 0, s_2 = 1\}$ overlap, while $\{s_1 = 1, s_2 = 0\}$ and $\{s_1 = 1, s_2 = 1\}$ also overlap. The result is, as seen in the plot, the overlap of the eigenvalues, this implies that the two that overlap share, at least in the domain where they overlap, the same energy level, which, ultimately, implies that these eigenvalues are degenerate in some subdomain, which, as we can see, is discontinuous.

A degenerate energy level implies the existence of a linear dependence. Now, the existence of these degeneracies also implies the neediness of a new quantum number that must be introduced to identify the state clearly. This quantum number is beyond our current study and further investigation is needed. The question that arises is whether this is actually a degeneracy since there is only degeneracy in a small subdomain, and this can be understood as simple crossings between the eigenvalues, meaning that in these crossings, via some phenomena, one is able to change from one state to another. If this interpretation and calculation are correct, and such phenomena exist, this implies that we can begin in a more favorable state and change to the other if in the domain of interests there is some advantage.

The plot draws attention to the fact that the eigenvalue for $\{s_1 = 0, s_2 = 0\}$ and $\{s_1 = 0, s_2 = 1\}$ are higher than $\{s_1 = 1, s_2 = 0\}$ and $\{s_1 = 1, s_2 = 1\}$. A fact that is impressive because in the single transmon we have the

contrary.

IV. CONCLUSIONS

The present work has helped to bring some clarity in the small field that are Trasmon qubits. We used a variety of methods, such as second quantization, matrix analysis, continuous diagonalization, and much more.

Using these methods, we obtained a differential equation that leads to the ground state wavefunction for the single transmon. Once we did this, we plotted our function together with the one that gives Scqubits just to see a good agreement between them. Afterwards, we focused our analysis on the eigenvalues of the system. Applying the sown method, consisting of continuous diagonalization, we obtained the searched eigenvalues, which were plotted together with a numerical solution. We could observe a good agreement between our approximations and the numerical solution. Specifically, between the upper states, since the eigenvalue for $n = 0$ was problematic using our method, leading us to apply a direct calculation using a Taylor expansion with $\cos(\varphi)$, which gave a good approximation, at least in a subdomain.

What continued was an analysis of a system of two coupled transmons. We performed a decoupling process just to see that once we did this, we could write the general formula in terms of the single transmon solution but with a change in the variables. It was interesting to compare it with a naively constructed solution via the direct product of the single transmon wavefunction, but changing the values. We observed that the coupling led to an asymmetry or a rupture of the symmetry between the probabilities of the state decomposition, a phenomenon that cannot be predicted via the direct product. It seems that the asymmetry has some relation with the energy E_{C_1} , since the higher E_{C_1} the higher the asymmetry.

From here, what lasted was to obtain the eigenvalues. Since we performed a decoupling procedure they would be the direct sum of the sub-spectrums, as said before. Constructing the functions and plotting them, we found that the behavior, meaning the form of the curve, resembles to the single transmon eigenvalues, which is a good sign, even though the umbral where the function is well behaved is higher.

Some interesting phenomena arose. It was the fact that we had an overlap in the eigenvalues, at least in a small subdomain. An interesting interpretation for this was to consider the existence of degeneracy, which requires a new quantum number. Another interpretation was that we only had crossings and we could take advantage of this, letting us interchange the states at these points via some mechanism.

All of this is beyond the scope of this current study and needs further investigation.

V. APPENDIX: SUPPLEMENTARY INFORMATION

A. Cooper Pairs

For electrons to acquire superconducting properties, a coherent movement joint with an attractive interaction between electrons is needed. For this to happen, the vibrations in the lattice allow the existence of an energy level below the Fermi energy, leading to the existence of a bound pair of coupled electrons.

This pair of coupled electrons are known as a Cooper pair. Electrons have spin $\hat{S} = -\frac{1}{2}$, and a spin coupling $l_{pair} = l_1 \oplus l_2 = [0, 1]$, meaning that the spin of the Cooper pair is an integer, implying that the Cooper pairs are bosonic particles, allowing us to define a bosonic space estate $H_B \in \mathbf{L}$. See Appendix [V B] for more information.

B. Space state

Taking into account the basic postulates of quantum mechanics, the state of a particle in a quantum system is described by a column vector residing in a complex Hilbert space denoted by $\mathbf{L}$. Considering a system made up of many particles, the state of each one of them is described by a subspace $\mathbf{L}_k \in \mathbf{L}$ of integral square functions in a set M_k . This space will be denoted as $\mathbf{L}^n$, where we can define an inner product such that

$$\langle f_1 | f_2 \rangle = (f_1, f_2) = \int f_1(\vec{x}) \overline{f_2(\vec{x})} d\vec{x}.$$

For the general state, the state that contains all the subspaces, it will be the direct sum of all the subspaces and the space that contains the vacuum vector, giving

$$\mathbf{L} = \sum_{n=0}^{\infty} \oplus \mathbf{L}^n.$$

There exist two fundamental subspaces of $\mathbf{L}$. The subspace of the fermionic particles $H_F \subset \mathbf{L}$ and the subspace of bosonic particles $H_B \subset \mathbf{L}$.

C. Creation and annihilation operators

Given a Hilbert space $\mathbf{L}$ with involution, if we can define another Hilbert space H_B , there exists a morfism between these two, such that $\mathbf{L} \iff H_B$, meaning that $\forall f \in \mathbf{L}$ we can define an operator $B(f) \in H_B$. Considering the spaces $\mathbf{L}$, H_B and the following matrices that belong to the space $\mathbf{L}$:

$$\hat{a}(\xi) = \begin{pmatrix} 0 & \delta(y, \xi) & 0 & 0 & \dots \\ 0 & 0 & \sqrt{2}\delta(x_1, y_1)\delta(y_2, \xi) & 0 & \dots \\ 0 & 0 & 0 & \sqrt{3}\delta(x_1, y_1)\delta(x_2, y_3)\delta(y_3, \xi) & \dots \\ \dots & \dots & \dots & \dots & \dots \end{pmatrix} \quad (24)$$

$$\hat{a}^\dagger(\xi) = \begin{pmatrix} 0 & 0 & 0 & 0 & \dots \\ \delta(y, \xi) & 0 & 0 & 0 & \dots \\ 0 & \sqrt{2}\delta(x_1, y_1)\delta(y_2, \xi) & 0 & 0 & \dots \\ 0 & 0 & \sqrt{3}\delta(x_1, y_1)\delta(x_2, y_3)\delta(y_3, \xi) & 0 & \dots \\ \dots & \dots & \dots & \dots & \dots \end{pmatrix}. \quad (25)$$

Denoting by P_B the projector that goes from $\mathbf{L} \rightarrow H_B$, we are able to define the operators

$$\hat{a}_B = P_B \hat{a}(\xi) P_B$$

$$\hat{a}_B^\dagger = P_B \hat{a}^\dagger(\xi) P_B$$

that represent the annihilation and creation operators of the Bose subspace. These operators verify the usual commutation relations, such that

$$[\hat{a}(f_1), \hat{a}(f_2)] = [\hat{a}^\dagger(f_1), \hat{a}^\dagger(f_2)] = 0$$

$$[\hat{a}(f_1), \hat{a}^\dagger(f_2)] = (f_1, f_2^\dagger).$$

D. Canonical transformation

Every unitary transformation in the space of states induces an automorphism such that $\hat{A} \iff \hat{U}\hat{A}\hat{U}^{-1}$. Considering the space H_B , where the annihilation and creation operators are defined, $\hat{a}_B$ and $\hat{a}_B^\dagger$, respectively, we can apply a unitary transformation so that we end up with $\hat{H}_B \iff \hat{U}H_B\hat{U}^{-1}$, meaning that we can define the transformed (rotated) annihilation and creation operators as:

$$\hat{b}^\dagger = \hat{U}\hat{a}^\dagger\hat{U}^{-1}$$

$$\hat{b} = \hat{U}\hat{a}\hat{U}^{-1}$$

that must obey the usual commutation relation

$$[\hat{b}^\dagger, \hat{b}] = 1.$$

In the case of the transmon qubit, given that we have two operators $\hat{\varphi}$ and $\hat{n}$ that fulfill the commutation relation $[\hat{\varphi}, \hat{n}] = i$, which is analogous to the one that verify the creation and annihilation operators, $[\hat{a}^\dagger, \hat{a}] =$

1, it implies that there exists a canonical transformation between these two variables.

We are able to write:

$$\phi = \alpha_1 a + \beta_1 a^\dagger$$

$$n = \alpha_2 a + \beta_2 a^\dagger,$$

applying this transformation to the commutation relation it gives

$$[\phi, n] = [\alpha_1 a + \beta_1 a^\dagger, \alpha_2 a + \beta_2 a^\dagger] = [\alpha_1, \alpha_2 a] + [\alpha_1 a, \beta_2 a^\dagger] + [\beta_1 a^\dagger, \alpha_2 a] + [\beta_1 a^\dagger, \beta_2 a^\dagger]$$

$$= [\alpha_1 a, \beta_2 a^\dagger] + [\beta_1 a^\dagger, \alpha_2 a] = i \rightarrow ([\alpha_1 a, \beta_2 a^\dagger] + [\beta_1 a^\dagger, \alpha_2 a]) \frac{1}{i} = 1$$

which give a relation for the coefficients

$$(\alpha_1 \beta_2 - \alpha_2 \beta_1) \frac{1}{i} = 1.$$

applying to the other commutators, we have

$$[\phi, \phi] = 0 = [\alpha_1 a + \beta_1 a^\dagger, \alpha_1 a + \beta_1 a^\dagger] = [\alpha_1 a, \beta_1 a^\dagger] + [\beta_1 a^\dagger, \alpha_1 a] = \alpha_1 \beta_1 - \alpha_1 \beta_1 = 0$$

and

$$[n, n] = 0 = [\alpha_2 a + \beta_2 a^\dagger, \alpha_2 a + \beta_2 a^\dagger] = \alpha_2 \beta_2 - \alpha_2 \beta_2 = 0$$

giving the system

$$(\alpha_1 \beta_2 - \alpha_2 \beta_1) \frac{1}{i} = 1$$

$$\alpha_1 \beta_1 - \alpha_1 \beta_1 = 0$$

$$\alpha_2 \beta_2 - \alpha_2 \beta_2 = 0$$

where we find that

$$\alpha_1 = \beta_1 = \frac{1}{\sqrt{2}}$$

$$\alpha_2 = -\beta_2 = \frac{1}{i\sqrt{2}}.$$

Re-escalating the operators as shown in Ref. [9], gives

$$\hat{\phi} = 2\sqrt{\eta} (\hat{a} + \hat{a}^\dagger)$$

$$\hat{n} - n_g = \frac{(\hat{a} - \hat{a}^\dagger)}{4i\sqrt{\eta}}, \quad (26)$$

which is a proper canonical transformation because the number of degrees of freedom are finite, and where we have defined $\eta = \sqrt{\frac{E_c}{8E_J}}$.

E. Functionals and derivative functionals

Following the steps of Ref. [1], given the space state $\mathbb{L}$, let us define an operator $\hat{A} \in \mathbb{L}$, which, following the same procedure as Ref. [10], accepts a matrix representation $\hat{A} = [A_{ij}]_{ij}$, where we can understand each entry of the matrix $[A_{ij}]_{ij}$ as $A_{ij} : \mathbb{L}^j \rightarrow \mathbb{L}^i$.

Each entry A_{ij} , can be defined by a generalized function, such that $A_{ij} = K_{ij}((x; j), (y; i))$.

Let us suppose the existence of a linear functional operator $\hat{B} : D \rightarrow \mathfrak{S}$, that verifies

1. $\hat{B}$ is dense in $D \subset \mathbb{L}$: $[B] = D$.

2. $\hat{B}$ is lineal and homogeneous: $\hat{B}(\alpha_1 f_1 + \alpha_2 f_2) = \alpha_1 \hat{B}(f_1) + \alpha_2 \hat{B}(f_2)$.

We know that $\forall f \in \mathbb{L} \exists \hat{B}(f) \in \mathbb{L} : B(f) = \int B(x) f(x)$. If we now consider the vacuum vector

$$|0\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \hat{F}_0,$$

taking into account the annihilation operator $\hat{a}$, it verifies $\hat{a}|0\rangle = 0$. Conversely, if the vector $|n\rangle$ is desired, the creation operator must be applied, giving

$$|n\rangle = \sum_{n=0}^{\infty} \frac{1}{\sqrt{n!}} \int K_n(\vec{x}_n) \prod_{n=1}^n \hat{a}^\dagger(\vec{x}_n) d^n x |0\rangle.$$

Consider now the functional derivatives respect the creation and annihilation operators. This implies

$$\frac{\delta}{\delta a(x)} |n\rangle = 0$$

$$\frac{\delta}{\delta a^\dagger(x)} |n\rangle = \sum_{n=0}^{\infty} n \frac{1}{\sqrt{n!}} \int K_n(\vec{x}_{n-1}, x) \prod_{n=1}^{n-1} \hat{a}^\dagger(\vec{x}_n) d^{n-1} x |0\rangle.$$

Having in mind how the operators $\hat{a}$ and $\hat{a}^\dagger$ act on $|n\rangle$, we can identify, for this case,

$$\hat{a}|n\rangle = \frac{\delta}{\delta a^\dagger}|n\rangle \quad (27)$$

$$\hat{a}^\dagger|n\rangle = a^\dagger|n\rangle. \quad (28)$$

F. Analytical diagonalization

Following the steps of Ref. [10]. Given the existence of an arbitrary operator $\hat{H}_{Tr} : H \rightarrow H$, this implies the existence of two subspaces H_1 and H_2 , such that $\hat{H}_{Tr} : R \subset H_1 \rightarrow D \subset H_2$. In the same way, given the existence of this, it implies the existence of a body $\mathfrak{S}$ where the subspaces H_1 and H_2 are defined, forming all of them, jointly with an intern and external operation, a bigger vector space $\mathbf{L}(H_1, H_2)$, known as the subspace of all the linear operators. We can also define $H_{m \times n}$ as the space containing all $m \times n$ matrices, meaning that $H_{m \times n} = \{H(m, n)\}$ that accepts a matrix representations $H_{Tr} \in H_{m \times n}$.

Given the operator $\hat{H}_{Tr} \in \mathbf{L}(H_1, H_2)$, we know that there is an isomorphism that links $\mathbf{L}(H_1, H_2)$ to $M_{m \times n}$ and vice versa, such that $H_{Tr} = \sum_{i=0}^m \left(\sum_{j=1}^n \alpha_j H_{Tr;i,j} \right) \beta_i$. This implies that the operator $\hat{H}_{Tr}$ can be written in matrix form, which means that there exists the matrix representation $H_{Tr} \in M_{m \times n}$.

Suppose the existence of an arbitrary domain $D = [a, b] \in H$, where H is a simply connected domain such that $H_{Tr;i,j} \in D$. If $\lambda_{H_{Tr}}$ represents an eigenvalue of H_{Tr} , we can also assume that $\forall \lambda_{H_{Tr}} \in D \rightarrow \lambda_{H_{Tr}} \in \mathfrak{R}$. This implies that $\exists H_1 \subset H : D \in H_1$ where $\lambda_{H_{Tr}} \in H_1$ are analytic.

Furthermore, assume $\det(H_{Tr}) = 0$ and the existence of a point $x = \alpha \in D$ which implies $\exists H_3 \subset H : \exists [\hat{a}, \hat{b}] \subset [a, b], [\hat{a}, \hat{b}] \in H_3, \alpha \in H_3$.

For definition, we see that the operator $\hat{H}_{Tr}$, with matrix representation H_{Tr} , is an endomorphism that operates in a space and gives a result in the same space, meaning that $\hat{H}_{Tr} : H \rightarrow H$. Given the fact that we have supposed the existence of eigenvalues for the operator, this implies that there exists a set of eigenfunctions, $\{h\}$, that will form a basis for the space $H_{3;m,n}$ that contains the matrix H_{Tr} .

Given the fact of the existence of $\{h\}$, it implies that the diagonalization of H_{Tr} is possible. Meaning that $\exists D_{H_{Tr}} : D_{H_{Tr}} \sim H_{Tr}$, which can be read as that exists a diagonal matrix, $D_{H_{Tr}}$, that is equivalent to a matrix form of the operator H_{Tr} . Where both of this represent the same endomorphism.

These two equivalent matrices must be equivalent by columns and rows. If $D_{H_{Tr}} \sim H_{Tr}$ by rows, this means that $\exists Q \in H_m : D_{H_{Tr};rows} = Q H_{Tr}$. If $D_{H_{Tr}} \sim H_{Tr}$ by columns, this means that $\exists P \in H_n : D_{H_{Tr};columns} = P H_{Tr}$. And if we take into account both of them, it implies that $D_{H_{Tr}} = Q H_{Tr} P$.

This implies [10] $\exists L, R : D_{H_{Tr}} = L H_{Tr} R = \text{diag}\{\vec{\lambda}_{H_{Tr}}\}$, which is the Smith form.

We will write here the same method of construction for the matrices L and R that is in Ref. [10].

Take the normal representation of a matrix, such that $A(x) = (a_{jk}(x))_{jk}$. If, by rotation, we impose that the element a_{11}^1 belongs to the maximal class, $\forall a_{j1}^1 \exists H_{j1}$ and $b_{j1}^1 : a_{j1}^1 = a_{11}^1 b_{j1}^1$. Analogously, $\forall a_{1j}^1 \exists H_{1j}$ and $b_{1j}^1 : a_{1j}^1 = a_{11}^1 b_{1j}^1$.

We can define the matrix E

$$E_{pq}^{jk}(\beta) = \delta_{pq} + \beta \delta_{p,j} \delta_{q,k}.$$

With this matrix, we can define the L and R matrices mentioned above, having

$$L = \left[\prod_{k=0}^{k=n-2} E_{n-k,1}^{p,q}(-b_{n-j,1}^1) \right] E_{n,1}^{p,q}(-b_{j1}^1) \quad (29)$$

$$R = E_{1,n}^{p,q}(-b_{in}^1) \left[\prod_{k=0}^{k=n-2} E_{1,n-k}^{p,q}(-b_{1,n-k}^1) \right], \quad (30)$$

which will ensure that

$$L A R = \text{diag}\{\vec{\lambda}_A\}. \quad (31)$$

1. Application for a single transmon

Suppose the existence of a matrix representation, $H_{Tr,lj}$, of the transmon Hamiltonian $\hat{H}_{Tr}$. Given the basic postulates of quantum mechanics, the eigenvalues of the system verify $\forall l \rightarrow \lambda_{H_{Tr};l} \in \mathfrak{R}$. This automatically implies that $\exists H_1 \subset H : \lambda_{H_{Tr};l} \forall l$ are analytic, as said in Ref. [10]. As shown, for going any further we need that $\det|H_{Tr}| = 0$, which is also automatically fulfilled by the fact that truncation is assumed.

The operator $\hat{H}_{Tr}$ is a Hamiltonian, meaning that $\hat{H}_{Tr} \in \mathbf{L} : \mathbf{L} \rightarrow \mathbf{L}$, which is an endomorphism, where $\mathbf{L}$ represents the space state, for instance $\exists L, R : D_{H_{Tr}} = L H_{Tr} R$. What last is to construct, iteratively, the matrices L and R . Given (8), $\exists h_{Tr;j1}^1 \in L : H_{Tr;j1}^1 = H_{Tr;11}^1 h_{Tr;j1}^1$ and $\exists h_{Tr;1j}^1 \in R : H_{Tr;1j}^1 = H_{Tr;11}^1 h_{Tr;1j}^1$. This will allow us to define the matrices $E(-h_{Tr;j1}^1)_{pq}^{jk}$ and $E(-h_{Tr;1j}^1)_{pq}^{jk}$ [10], such that

$$\begin{aligned} E(-h_{Tr;i1})_{pq}^{jk} &= \delta_{pq} - h_{Tr;i1} \delta_{pj} \delta_{qk} \\ E(-h_{Tr;1i})_{pq}^{jk} &= \delta_{pq} - h_{Tr;1i} \delta_{pj} \delta_{qk}. \end{aligned}$$

We need to determine $h_{Tr;i1}$ and $h_{Tr;1i}$. But by definition, we have that

$$h_{Tr;i1} = \frac{H_{Tr;i1}^1}{H_{Tr;i1}}, \quad h_{Tr;1i} = \frac{H_{Tr;1i}^1}{H_{Tr;1i}},$$

implying

$$\begin{aligned} L(-h_{Tr;i1})_{pq;i}^{jk} &= \delta_{pq} - \frac{H_{Tr;i1}^1}{H_{Tr;i1}} \delta_{pj} \delta_{qk} \\ R(-h_{Tr;1i})_{pq;i}^{jk} &= \delta_{pq} - \frac{H_{Tr;1i}^1}{H_{Tr;1i}} \delta_{pj} \delta_{qk}. \end{aligned}$$

G. Group theory and $O^+(2)$ group

As in Ref. [17], we define a group G as a collection of objects, meaning that $G = \{g_i(\vec{\alpha})\}$ verifies

- $\exists \cdot_v : G \times G \rightarrow G$.
- $g_i(g_j g_k) = (g_i g_j) g_k \quad \forall i, j, k$.
- $\exists g_0 : g_i g_0 = g_0 g_i = g_i$.
- $\exists g_i^{-1} : g_i g_i^{-1} = g_0$.

Which can be summarized as a closed set around the product operation.

Given G , let us suppose the existence of a group of linear operators $T \in R : T = \{\hat{T}_{g_i}\}$ that is homomorphic to G . This means that T constitutes a representation of G . Which means that $\hat{T}_{g_i} \hat{T}_{g_j} = \hat{T}_{g_i g_j}$ holds.

Given R , if the collection $\{\hat{e}_i\}$ constitutes a basis of R , $\forall x \in R \rightarrow x = \sum_{i=0}^n x_i \hat{e}_i$. Now, we define $\hat{T}_{g_i}$, but it will be defined if we define its behavior on the basis $\{\hat{e}_i\}$. Let us write

$$\hat{T}_{g_i} e_k = \sum_{r=1}^n D_{rk}(g_i) e_r,$$

where we can see that $\forall g_i \in G \quad \exists |D_{rk}(g_i)| : D(g_i) D(g_j) = D(g_i g_j)$. That is, $D(g_i)$ constitutes a representation of G .

Let us suppose that the elements of G verify the relation

$$g(\vec{\alpha}) g(\vec{\alpha}') = g(\vec{\alpha}'') \quad \vec{\alpha} = \vec{\phi}(\vec{\alpha}; \vec{\alpha}'),$$

meaning that we are dealing with a continuous Lie group, and if we have a finite dimension we will be dealing with compact groups.

Given the existence of G , let us suppose that is a uniparametric group where its elements are differentiable functions of the parameter. Let us further suppose that the parameter is defined in a way that

$$g(\alpha_1) g(\alpha_2) = g(\alpha_1 + \alpha_2) \quad g(0) = g_0,$$

differentiating by α_1 and then do $\alpha_1 = 0, \alpha_2 = \alpha$, we will end up with $\frac{dg}{d\alpha} = g(\alpha) \left[\frac{dg}{d\alpha} \right]_{\alpha=0}$, that will have a solution of the form $g(\alpha) = e^{I_\alpha \alpha}$. Here, I_α are the infinitesimal matrices of the group.

Now, let us analyze $O^+(2)$. This group consists of rotations in space. Given that this group is abelian, it will be uniparametric. Hence, for every irreducible representation we will put in correspondence a number that must verify

$$\Xi(\alpha_1) \Xi(\alpha_2) = \Xi(\alpha_1 + \alpha_2) \quad \Xi(0) = \Xi(2\pi),$$

where it follows that $\Xi(\alpha) = e^{im\alpha}$. Hence, we can identify the group as

$$O^+(2) = \begin{pmatrix} \cos(\alpha) & -\sin(\alpha) \\ \sin(\alpha) & \cos(\alpha) \end{pmatrix}. \quad (32)$$

H. Existence of a Ground-State wavefunction for the two coupled transmons

The existence of a proper canonical transformation verifies the existence of a vacuum vector $|0_1 0_2\rangle$ and given the fact that we are now dealing with a decoupled system and knowing that the variable associated to the basis with which we are dealing, the number of Cooper pairs, are considered bosonic particles, as shown in Ref. [12], they will verify that $|0_1 0_2\rangle = |0_2 0_1\rangle$, which means that we can write $|0_1 0_2\rangle = \frac{1}{\sqrt{2}}(|0_1 0_2\rangle + |0_2 0_1\rangle)$, and further $|0_1 0_2\rangle = \frac{1}{\sqrt{2}}(|0_1\rangle|0_2\rangle + |0_2\rangle|0_1\rangle)$.

Consider the product operator $\hat{a}_1 \hat{a}_2$, operating above $|0_1 0_2\rangle$, which is the vacuum vector. Meaning that $\hat{a}_1 \hat{a}_2 |0_1 0_2\rangle = 0$.

$$\hat{a}_1 \hat{a}_2 |0_1 0_2\rangle = \frac{\hat{a}_1 \hat{a}_2}{\sqrt{2}} (|0_1\rangle|0_2\rangle + |0_2\rangle|0_1\rangle) = 0.$$

By definition $[\hat{a}_1, \hat{a}_2] = 0$, which translates into

$$(\hat{a}_2 \hat{a}_1 |0_1\rangle|0_2\rangle + \hat{a}_1 \hat{a}_2 |0_2\rangle|0_1\rangle) = 0.$$

Since $\hat{a}_{2,1}$ does not interfere with $|0_{1,2}\rangle$, we can write

$$(\hat{a}_1|0_1\rangle\hat{a}_2|0_2\rangle + \hat{a}_2|0_2\rangle\hat{a}_1|0_1\rangle) = 0.$$

This allows

$$(\hat{a}_1|0_1\rangle\hat{a}_2|0_2\rangle) = 0.$$

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